

Vectors

1 Introduction

These problems are a review of vector notation and trigonometry. Recall that a two-dimensional vector requires two pieces of information to describe it. In Problem I, the two pieces of information are the magnitude and direction. In Problem II, the two pieces of information are the horizontal and vertical components. In each problem, you convert from one description to the other.

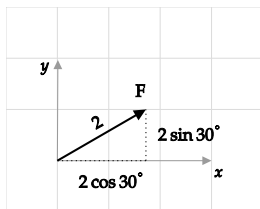
2 Problem I

A vector $\vec{\mathbf{F}}$ has a magnitude of 2 and makes an angle of 30° with the x -axis (with positive rotation counterclockwise).

1. Draw $\vec{\mathbf{F}}$.
2. Write $\vec{\mathbf{F}}$ in the form $F_x\hat{\mathbf{i}} + F_y\hat{\mathbf{j}}$.

Solution

- 1.



2. $\vec{\mathbf{F}} = 2 \cos(30^\circ)\hat{\mathbf{i}} + 2 \sin(30^\circ)\hat{\mathbf{j}} = \sqrt{3}\hat{\mathbf{i}} + \hat{\mathbf{j}}$, where $\cos(30^\circ) = \sqrt{3}/2$ and $\sin(30^\circ) = 1/2$ was used in the last step.

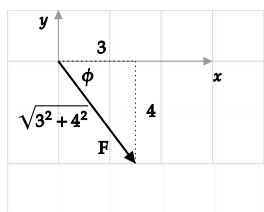
3 Problem II

Given $\vec{\mathbf{F}} = 3\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$,

1. Draw $\vec{\mathbf{F}}$.
2. Compute F .
3. The angle $\vec{\mathbf{F}}$ makes with respect to the x -axis (with positive rotation counterclockwise).

Solution

- 1.



$$2. F = \sqrt{3^2 + 4^2} = 5$$

$$3. \theta = 360^\circ - \phi = 360^\circ - \frac{180}{\pi} \tan^{-1}(4/3) \simeq 307^\circ$$